The Impact of South Levantine Early Bronze Age Communities On Their Landscapes

Thesis submitted in accordance with the requirements of
the University of Liverpool for the degree of Doctor in Philosophy
by
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September, 2009

In loving memory of

Burt Jorgenson

I miss you terribly Grandpa.

Abstract

Conventional views suggest that the Early Bronze Age I-III periods c. 3600-2400 BC saw the appearance of the first urban communities in the southern Levant. During this period we see the emergence of some larger communities, fortified sites, large public buildings, but not at the same scale as the regions of the northern Levant or southern Mesopotamia. This view conceives communities of this period as small-scale versions of the third millennium city-states in southern Mesopotamia and northern Syria (e.g. Ben-Tor (1992a)). Alternatively some authors now suggest that the communities of the S. Levant in this period are no more than 'corporate villages', but that developments still suggest significant degrees of modification of the landscape by these communities, indeed that for the first time communities 'inscribe' themselves on the landscape (Chesson 2000, Philip 2001).

By studying these phenomena using Geographical Information Science (GIS) a number of interesting issues arise. If we see the appearance of a significant number of larger communities for the first time, often embedded in a network of smaller settlements, what sort of impact might they be having upon their landscape? Should we see significant degrees of environmental modification, even negative impacts? Do these larger communities require levels of surplus production by other communities, therefore allowing us to understand them as urban? How much do distinctive land management strategies and mortuary monuments suggest that the landscape was laid out in a new way, or more specifically, in a way that allowed communities and/or their elites to stamp their identity on the landscape?

A number of specific case studies can be integrated into a GIS which will allow these issues to be addressed. To make this model work, case studies will be chosen in which the excavated site(s) yields enough applicable data and also has some evidence of interaction with other settlements in the surrounding landscape. This will allow an approach to alternative reconstructions of population size and subsistence needs. One case study could be the excavations at Tell esh Shuna, an EB I site in the Northern Jordan valley; another focus area might be the Hula valley in Israel. In these areas extant dolmen fields exist which are contemporary with some of these settlements allowing reconstruction of other aspects of the landscape. From the excavated environmental, palaeobotanical, and faunal evidence one can look at the past palaeo-environments and land management strategies.

Using various soil and topographic data the GIS will allow the reconstruction of alternative possible land management arrangements across given landscapes and possible environmental impacts. Furthermore, this model will be capable of addressing a number of other issues including ritual/mortuary land connections.

Acknowledgements

As I sit here, at the end of this amazing journey which began so many years ago, I realise when looking back just how many people have had a part in getting to this point. Knowing that an attempt to include all of you in this short space would surely result in an incomplete list, I want to say thank you to everyone. The fact that you are reading this probably means that you are one of these people.

I first have to say a massive thank you to Dr. Douglas Baird for his guidance and support over the last four years. Doug has been an extraordinary mentor and role model, and I consider myself very privileged indeed to have had him as a supervisor. Likewise, I would also like to thank Dr. Matthew Fitzjohn for his advice, support and encouragement; in addition to the help he provided in his supervisory role, there were many times over the last four years when he somehow sensed when I was down, and knew what to say, and his words would pick me up and motivate me to carry on.

The support of my colleagues has meant so much to me. Dr. Nick Taylor has really been there for me, both as a colleague, and a friend and has become a much loved part of our family. My office mates, and good friends, Adnan Baysal and Emma Twigger, are also owed many thanks; I am sure there are times that I must have been on the verge of driving them mad. I very much enjoyed sharing office space with them, and very much appreciate the many insights, suggestions and encouragement that they gave. Dr. Samantha Hemsley and Dr. Dana Campbell 'adopted' me when I first arrived at Liverpool, and have become good friends whose help and support over the course of this journey I have very much appreciated. Lyn Hughes has likewise been so good to me, reading and commenting on my work; I want to thank her for that, as well as for her support, encouragement, honesty and much valued friendship. I must also thank Tim Sutton of South Africa for teaching me C++ programming; his many hours of much appreciated help (via internet chat) made it possible for me to include many important features in the software I created for this research (Landuse Analyst) that would otherwise have had to be left out. To all my other 'University' friends and colleagues, I offer my most sincere thank you, for they have all been an important part of this journey.

I must also express my gratitude to Dr. Shawn Bubel, whom I very much admire and respect, for giving me the 'archaeology bug' and inspiring me to continue on with my education. Dr. Roy Wilson played a big part in 'keeping me on track' when I first began my undergraduate degree, and his continued support over the years has meant very much to me. Mr. Peter White has been a dependable mentor for me, and I must offer my thanks to him for all of his guidance, support, encouragement and words of wisdom over the last 20 years. *Nihil illigitimus carborundum!*

Leaving 'home' in Alberta and coming to Liverpool with my wife and three children (all under the age of eight) to pursue a Ph.D. has certainly proved challenging. The help and support of our circle of friends outside the University has been incredibly important. Robert Martin and Bob Clarke have simply become family, and I cannot imagine how we could have gotten through this without them both. Alan Riley has also been a steadfast pillar of friendship and support for which I will always be grateful. Peter Crossan is a respected friend, who was there with me when I once stood at an important crossroads on this journey, and for this I am extremely grateful.

Terry and Velma Pearen are two very wonderful people that have played a big role not only in the completion of this thesis, but in my life in general. Their constant support and encouragement has meant more than they can imagine, and I will forever be grateful.

I would like to thank my ‘Fairy Godmother’, Georgia Sullivan, for her unwavering support. She has, without exception, been there for me, and for this I am eternally grateful.

It was not easy for our families in Canada to be supportive of a decision that would result in us being separated from them for such a long time. In spite of this, they have. Although difficult at times, I know, their love and support made this possible. Wayne and Marguerite, I cannot tell you how much I appreciate your help. You have been nothing short of amazing. Mom and Dad, you sacrificed so much for us kids as we were growing up. Everything about ‘me’ which has made it possible to get where I am I attribute to the two of you. You are wonderful parents, and I love and respect you both very much. I hope I’ve done you proud.

The most important thank you I can offer, however, is to my family. Jared, Brooklyn and Kaileigh, you will never know how much your love and support has meant to me. Over the last four years I have not been able to spend the time with you that you have deserved. Despite everything, each of you have excelled at everything you do, and I am incredibly proud of each one of you. I just hope that someday, this sacrifice might in some small way inspire each of you to the greatness I know each of you is destined for. Always follow your hearts kids, and never let anyone tell you that your dreams are impossible.

Finally, I have the impossible task of trying to come up with the words that will express to my wonderful wife just how much.... I don’t even know how to begin. Consolation from my inability to find these words does come, however, when I look into her beautiful eyes, for it is then that I know ‘she knows’. Rae, you are my love. My life. My forever. *Thank you*.

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